

► Conclusion

In this paper, we have presented the methodology used to develop a new global index of occupational exposure to GenAI, as well as new estimates of the global effects on employment. We build on the approach developed by the ILO in 2023, but with several notable refinements. First, by beginning our analysis with the Polish 6-digit occupational structure, we expand by ten-fold the amount of detailed task information available for the analysis of task automation. In addition, using a representative sample from the 29,753 tasks in the Polish occupational classification system and a survey of 1,640 people in Poland, we captured the opinions of each 1-digit ISCO-08 group on the potential for task automation in the job categories in which they are currently employed (52,558 data points). We then compared this input with an expert survey and conducted subsequent rounds of revisions among a smaller international expert group for the tasks with the highest degree of disagreement. Based on this process, we built a repository of knowledge with adjusted task scores that goes beyond Poland's national specificities and reflects a broader consensus on the task automation potential of current GenAI tools.

We subsequently used this repository to develop an AI assistant that could predict scores for these tasks based on this additional information as well as the technical documentation of ISCO-08. These scores update the original set of scores produced by the ILO in 2023 (Gmyrek et al., 2023). The use of a verified “knowledge base” presents an important methodological improvement, which greatly enhances stability of our predictions across independent LLMs (see Section 6), and which can be used to produce similar score predictions for other occupational classification systems that contain job-specific tasks. In addition, we make all task-level scores publicly available, which also makes it possible for researchers to organize the occupational exposure in different ways than our proposed framework of exposure gradients.

The new global index presented in this paper is generic, due to the nature of ISCO-08 and the structure of tasks associated with occupations in this framework. Its definite strength – similar to ISCO-08 – is that it provides a common global denominator and a direct link to national labour forces surveys held in the ILO harmonized micro data repository, which relies on the same classification of occupational structures. Even though all countries do not use ISCO-08, mappings of the national systems to this common international standard are available. This eliminates the need for crosswalks from other national systems (e.g. O*NET), which involves tenuous assumptions about the similarity of detailed occupations in the technologically advanced context of the USA to other countries, especially non-HICs. The simple and comparable character of our ISCO-08 based index enables direct production of global, regional and income-based employment estimates, based on the same statistical modelling techniques that are used by the ILO for other global figures (see Section 11).

Our 2025 scores are presented in a revised occupational classification system, adopting a framework of four progressively increasing exposure gradients, that better highlights the importance of potential AI exposure and task variability for determining possible outcomes. We produced a new set of global estimates of GenAI exposure based on this revised classification.

The new index maintains many of the headline findings of the 2023 methodology. Clerical occupations continue to have the highest exposure levels, albeit with some important downward revisions to individual tasks scores. Such occupations include data entry clerks, typists, accounting, bookkeeping clerks and administrative secretaries. Additionally, some occupations – such as financial analysts, and web and multimedia developers (gradient 4), as well as bank tellers, translators, applications programmers, investment advisers (gradient 3) – have increased their

levels of exposure, highlighting the expanding abilities of GenAI technology to potentially automate more specialized tasks in professional and technical roles.

Overall, the revised global employment estimates of GenAI exposure are lower than in our 2023 paper, though still, one in four jobs globally (24%) has some degree of exposure, and one in three jobs in high-income countries. What is particularly striking is the continued association of gender with occupations with higher levels of exposure, especially in higher-income countries. Compared to 7.8% of female employment and 2.9% of male employment in high-income countries classified as having high automation potential in 2023, the current study increases these figures to 9.6% for females and 3.5% for males, respectively.

Notwithstanding these findings, we stress that such exposure does not imply the immediate automation of an entire occupation, but rather the potential for a large share of its current tasks to be performed using this technology. Whether this leads to the disappearance of an occupation or workforce replacement is a more complex question – one that will depend on the initial decision to adopt the technology, but also the extent to which individuals in these occupations are given opportunities to learn to work with these technologies and adapt to the evolving nature of their tasks. As argued by numerous economists, the key to unlocking the productivity benefits of GenAI lies not in the search of outright labour savings, but in the extent to which human expertise can be complemented with new technological capacities to reshape occupations and generate additional value-added (Acemoglu, 2024; Autor, 2024).

Ensuring such positive synergies necessitates an approach to technology's adoption at the workplace that is human-centered. With one in four workers across the world in an occupation with some degree of GenAI exposure, there is need to ensure that the transition is managed through social dialogue, both to minimize “technological unemployment” and the hardship associated with it, but also to ensure that for the majority who remain employed, the technology is implemented in a way that supports them in carrying out their jobs. Workers know their jobs best and can play an important role in the design, adaptation and use of the technology at the workplace, to the benefit of both working conditions and productivity.

Annex

1. Exposure by 4-digit ISCO-08 occupation

► Table A1. ISCO-08 occupations by exposure gradient

Exposure	4-digit code	Occupation Name	Mean	SD
Gradient 4	4132	Data Entry Clerks	0.7	0.03
Gradient 4	4131	Typists and Word Processing Operators	0.65	0.05
Gradient 4	4311	Accounting and Bookkeeping Clerks	0.64	0.07
Gradient 4	4312	Statistical, Finance and Insurance Clerks	0.64	0.02
Gradient 4	3311	Securities and Finance Dealers and Brokers	0.63	0.04
Gradient 4	4419	Clerical Support Workers Not Elsewhere Classified	0.63	0.03
Gradient 4	2413	Financial Analysts	0.62	0.06
Gradient 4	4313	Payroll Clerks	0.61	0.08
Gradient 4	5244	Contact Centre Salespersons	0.61	0.1
Gradient 4	2513	Web and Multimedia Developers	0.6	0.08
Gradient 4	3312	Credit and Loans Officers	0.6	0.04
Gradient 4	4110	General Office Clerks	0.6	0.1
Gradient 4	4416	Personnel Clerks	0.6	0.09
Gradient 3	2643	Translators, Interpreters and Other Linguists	0.59	0.11
Gradient 3	4120	Secretaries (general)	0.58	0.1
Gradient 3	4211	Bank Tellers and Related Clerks	0.58	0.09
Gradient 3	4222	Contact Centre Information Clerks	0.58	0.06
Gradient 3	4414	Scribes and Related Workers	0.58	0.08
Gradient 3	2412	Financial and Investment Advisers	0.57	0.11
Gradient 3	2514	Applications Programmers	0.57	0.08
Gradient 3	2521	Database Designers and Administrators	0.57	0.08
Gradient 3	2522	Systems Administrators	0.57	0.05
Gradient 3	3314	Statistical, Mathematical and Related Associate Professionals	0.57	0.1
Gradient 3	3331	Clearing and Forwarding Agents	0.57	0.09
Gradient 3	4225	Inquiry Clerks	0.57	0.09
Gradient 3	4226	Receptionists (general)	0.57	0.17
Gradient 3	2120	Mathematicians, Actuaries and Statisticians	0.56	0.09
Gradient 3	4221	Travel Consultants and Clerks	0.56	0.07
Gradient 3	2431	Advertising and Marketing Professionals	0.55	0.09
Gradient 3	2519	Software and Applications Developers and Analysts Not Elsewhere Classified	0.55	0.16
Gradient 3	2622	Librarians and Related Information Professionals	0.55	0.04

Exposure	4-digit code	Occupation Name	Mean	SD
Gradient 3	2631	Economists	0.55	0.08
Gradient 3	2641	Authors and Related Writers	0.55	0.09
Gradient 3	3342	Legal Secretaries	0.55	0.13
Gradient 3	4227	Survey and Market Research Interviewers	0.55	0.16
Gradient 3	2112	Meteorologists	0.54	0.13
Gradient 3	2642	Journalists	0.54	0.1
Gradient 3	3343	Administrative and Executive Secretaries	0.54	0.14
Gradient 3	4223	Telephone Switchboard Operators	0.54	0.15
Gradient 3	2512	Software Developers	0.53	0.07
Gradient 3	3321	Insurance Representatives	0.53	0.12
Gradient 3	3344	Medical Secretaries	0.53	0.11
Gradient 3	3514	Web Technicians	0.53	0.12
Gradient 3	2523	Computer Network Professionals	0.52	0.06
Gradient 3	3252	Medical Records and Health Information Technicians	0.52	0.1
Gradient 3	2411	Accountants	0.51	0.08
Gradient 3	2434	Information and Communications Technology Sales Professionals	0.51	0.09
Gradient 3	4224	Hotel Receptionists	0.51	0.07
Gradient 3	4413	Coding, Proofreading and Related Clerks	0.51	0.2
Gradient 3	2433	Technical and Medical Sales Professionals (excluding ICT)	0.5	0.1
Gradient 3	4323	Transport Clerks	0.5	0.1
Gradient 2	2166	Graphic and Multimedia Designers	0.49	0.09
Gradient 2	2511	Systems Analysts	0.49	0.05
Gradient 2	2529	Database and Network Professionals Not Elsewhere Classified	0.49	0.06
Gradient 2	3313	Accounting Associate Professionals	0.49	0.13
Gradient 2	3322	Commercial Sales Representatives	0.49	0.08
Gradient 2	3352	Government Tax and Excise Officials	0.49	0.12
Gradient 2	2153	Telecommunications Engineers	0.48	0.12
Gradient 2	2632	Sociologists, Anthropologists and Related Professionals	0.48	0.12
Gradient 2	3332	Conference and Event Planners	0.48	0.14
Gradient 2	4213	Pawnbrokers and Money-lenders	0.48	0.06
Gradient 2	1346	Financial and Insurance Services Branch managers	0.47	0.12
Gradient 2	2356	Information Technology Trainers	0.47	0.1
Gradient 2	2621	Archivists and Curators	0.47	0.11
Gradient 2	2633	Philosophers, Historians and Political Scientists	0.47	0.06
Gradient 2	2656	Announcers on Radio, Television and Other Media	0.47	0.12
Gradient 2	3512	Information and Communications Technology User Support Technicians	0.47	0.19
Gradient 2	2421	Management and Organization Analysts	0.46	0.1
Gradient 2	4411	Library Clerks	0.46	0.18